

$$V e(t) = R_1 i_1(t) + L \frac{di_1(t)}{dt} + \frac{1}{C} \int [i_1(t) - i_2(t)] dt$$

$$\frac{1}{C} \int [i_1(t) - i_2(t)] dt = R_2 i_2(t)$$

$$V s(t) = R_2 i_2(t)$$

Transformada de Laplace:

$$V e(s) = R_1 I_1(s) + L s I_1(s) + \frac{I_1(s) - I_2(s)}{C s}$$

$$\frac{I_1(s) - I_2(s)}{C s} = R_2 I_2(s)$$

$$V s(s) = R_2 I_2(s)$$

Despejar $I_1(s)$:

$$\frac{I_1(s)}{C s} - \frac{I_2(s)}{C s} = R_2 I_2(s)$$

$$\frac{I_1(s)}{C s} = R_2 I_2(s) + \frac{I_2(s)}{C s}$$

$$\frac{I_1(s)}{Cs} = (R_2 + \frac{1}{Cs})I_2(s)$$

$$I_1(s) = (CR_2s + 1)I_2(s)$$

Sustituir $I_1(s)$:

$$Ve(s) = R_1(CR_2s + 1)I_2(s) + Ls(CR_2s + 1)I_2(s) + \frac{(CR_2s + 1)I_2(s) - I_2(s)}{Cs}$$

$$Ve(s) = [CR_1R_2s + R_1 + CR_2Ls^2 + Ls + \frac{CR_2s + 1}{Cs} - \frac{1}{Cs}]I_2$$

$$Ve = [CR_2Ls^2 + (CR_1R_2 + L)s + R_1 + R_2]I_2$$

Función de transferencia:

$$\frac{Vs(s)}{Ve(s)} = \frac{R_2I_2}{[CR_2Ls^2 + (CR_1R_2 + L)s + R_1 + R_2]I_2} = \frac{R_2}{CR_2Ls^2 + (CR_1R_2 + L)s + R_1 + R_2}$$

Ecuaciones integrodiferenciales:

$$R_1i_1(t) = Ve(t) - L\frac{di_1(t)}{dt} - \frac{1}{C}\int[i_1(t) - i_2(t)]dt$$

$$i_1(t) = [Ve(t) - L \frac{di_1(t)}{dt} - \frac{1}{C} \int [i_1(t) - i_2(t)] dt] \frac{1}{R_1}$$

$$i_2(t) = [\frac{1}{C} \int [i_1(t) - i_2(t)] dt] \frac{1}{R_2}$$

Error estacionario:

$$e(s) = \lim_{s \rightarrow 0} sVe(s) [1 - \frac{Ve(s)}{Ve(s)}]$$

$$e(s) = \lim_{s \rightarrow 0} (s \times \frac{1}{s}) [1 - \frac{R_2}{CR_2Ls^2 + (CR_1R_2 + L)s + R_1 + R_2}]$$

$$e(s) = 1 - \frac{R_2}{R_1 + R_2} = \frac{R_1 + R_2}{R_1 + R_2} - \frac{R_2}{R_1 + R_2} = \frac{R_1}{R_1 + R_2}$$

Control:

$$\frac{10}{40 + 10} = \frac{1}{5}$$

Caso:

$$\frac{80}{10 + 80} = \frac{8}{9}$$

Análisis de estabilidad de lazo abierto:

$$\partial_{1,2} = \frac{-(CR_1R_2 + L) \pm \sqrt{(CR_1R_2 + L)^2 - 4(CR_2Ls)(R_1 + R_2)}}{2(CR_2Ls)}$$

Control:

R1=10ohms, R2=40ohms, C=100uF, L=10uH

$$\partial_1 = -5.008 \times 10^7$$

$$\partial_2 = -2.496 \times 10^{10}$$

Caso:

R1=80ohms, R2=10ohms, C=20uF, L=10uF

$$\partial_1 = -7.9119 \times 10^7$$

$$\partial_2 = -7.280 \times 10^6$$

El sistema es estable dado que ambas raíces son negativas y reales entonces el sistema permite una respuesta sobreamortiguada